Drone Intrusion Detection System

PUSHPAK Grand Challenge 3: Security of Drones, Objective 2

Stage 2 -- Detailed Evidence-Supported IDS Report Template

This template is intended for the Stage 2 detailed submission. Teams are expected to go beyond the Stage 1 proof of concept and provide validation evidence from actual flight operation, on real and traceable data, that builds confidence the intrusion detection system can be deployed on a UAV and subsequently tested in flight.

Team and Submission Information

Particular	Details
Team ID	
Team Name	
Date of Submission	
IDS Name / Version	

Report Structure and Section Weightage

Section	Weight
1. Executive IDS Summary	5
2. Stage 1 Baseline and Design Evolution	8
3. Updated/implemented Requirements and Architecture	7

4. Integrated Methodology and Workflow	8
5. Detection Methodology and Attack Coverage	18
6. Data Sources, Features and Detection Analysis	12
7. False-Alarm Behaviour and Alerting	8
8. Validation, Testing and Reproducibility	14
9. Compute, Integration, Usability and Reporting	8
10. Final Deployable Package	7
11. References, Data Traceability and Integrity	5
Total	100

Submission note: Keep the main report concise and evidence-driven. Source code, test suites, flight logs and datasets, attack-injection scripts, ground-truth files, result files and the demonstration video must be included in the submitted archive; a link alone is not sufficient. Detailed results, extra plots and raw outputs may be attached as appendices. Results from actual flight operation carry full weight; results from simulated or generated data are treated as design evidence and carry limited weight. The emphasis is on traceable methodology, detectors that decide from telemetry and not from test labels, and results that come from the submitted code on the data described.

1. Executive IDS Summary [5 Marks]

Briefly summarise the final Stage 2 IDS, the major changes from Stage 1, and the level of confidence in the final result.

Parameter	Final Stage 2 Value	Unit	Evidence / Section Reference
IDS name and version		--	
Attack classes covered (GPS spoofing, MAVLink anomalies, command injection, telemetry manipulation, DoS, firmware integrity)		count	
Detection layers (rules, physics, statistics, learned model)		count	
Autopilot platforms tested (ArduPilot, PX4)		count	
Placement (onboard, companion computer, ground station, tap)		--	
Telemetry messages monitored		count	
Detection accuracy on data not used for tuning		%	
False-alarm rate on benign flights		per hour	
Benign flight time and distance tested		h / km	
Median and 95th-percentile detection latency		ms or s	

Target hardware used for resource measurement		--	
CPU use and memory on that hardware		% core / MB	
Tests included in the submitted archive / passing		count	
Real flight-log hours used		h	
Alert integrity mechanism (signing, hash chain)		--	
Runs fully offline?		Yes / No	

Top three reasons why the IDS is considered ready for extended flight trials:

1.

2.

3.

Top three remaining uncertainties or risks:

1.

2.

3.

2. Stage 1 Baseline and Design Evolution [8 Marks]

Do not repeat the complete Stage 1 report. Summarise the baseline, state what changed, and justify each change using Stage 2 testing, validation or engineering judgement.

Item	Stage 1 Value / Approach	Stage 2 Value / Approach	Change Made?	Reason / Evidence
Attack classes covered			Yes / No	
Detectors and layers			Yes / No	
Platforms and protocols			Yes / No	
Data used for validation			Yes / No	
Detection accuracy			Yes / No	
False-alarm rate and benign exposure			Yes / No	
Detection latency			Yes / No	
Compute footprint and hardware			Yes / No	
Alerting and logging			Yes / No	
Integration approach			Yes / No	
Architecture			Yes / No	

2.1 Stage 1 Claims Revisited

List each capability claimed or planned in Stage 1 and state whether it is now demonstrated.

Stage 1 Claim	Demonstrated / Partly / Withdrawn	Reason	Supporting Evidence

2.2 Stage 1 Open Questions Addressed

Include how the team interprets 'distance covered' and what it measured.

Open Question from Stage 1	How Addressed in Stage 2	Result / Evidence	Remaining Uncertainty

3. Updated/Implemented Requirements and Architecture [7 Marks]

Requirement / Target	Challenge Target	Final Team Target	Status / Comment
Attack coverage	Detect the six attack classes: GPS spoofing, MAVLink anomalies, command injection, telemetry manipulation, denial of service, firmware integrity		
Detection accuracy	High detection across attack scenarios, including subtle attacks		
False alarms	Low false-alarm rate on benign flight		
Latency	Timely alerts, fast enough to act on		
Compute	Low overhead on flight-compatible hardware		
Integration	Works with ArduPilot and PX4 and a ground station without degrading flight		
Alert security	Tamper-evident alerts and logs		
Design outcome	Validated, deployable IDS		

Describe the final IDS architecture, including data sources and tap point, pre-processing, feature extraction, detection layers, fusion, alerting, logging, interfaces and where the IDS runs. State how the IDS avoids interfering with flight control.

Insert final system architecture figure here

4. Integrated Methodology and Workflow [8 Marks]

This section should make the design process reproducible. Show how data sources, features, detectors, fusion, alerting and testing were linked. Include a logic-flow diagram.

Insert workflow / logic-flow diagram here (ingest, extract features, detect, fuse, alert, log, evaluate)

Sub-domain	Tool / Code / Method	Main Inputs	Main Outputs	How Verified / Checked
Data ingestion and parsing				
Feature extraction				
Detection layers				
Fusion and alert policy				
Alerting and logging				
Attack injection and test generation				
Metric computation				

Explain how results from one stage affect the others, for example how false-alarm results changed a threshold, or how a missed attack class changed a detector.

5. Detection Methodology and Attack Coverage [18 Marks]

This is the most heavily weighted section. Detectors must decide from telemetry and commands, not from test labels, injected-attack flags or the attack generator's state. Attacks should include subtle cases near normal noise levels, not only large constant offsets.

5.1 Methodology

Item	Details Required
Detectors per attack class	Name each detector, the layer it belongs to and the attack class it addresses
Inputs read by each detector	List every message and field each detector uses; confirm that none reads ground truth, attack flags or labels
Thresholds and training	How thresholds were set, any training data, and the split between calibration, validation and test data
Decision logic	Rules, statistics or model; persistence and fusion; what triggers an alert
Latency design	Where time is spent and what the minimum detection time is
Baseline comparison	A simple threshold or the autopilot's own checks, for comparison

Known blind spots	Attacks the IDS does not detect, and why
Limitations	Known limits of the methodology

5.2 Attack Coverage and Test Matrix

Attack class	Scenarios (count, magnitude range, including subtle)	Flight evidence used (flight log, live telemetry, replay)	Injection method and ground-truth source	Detected / total
GPS spoofing				
MAVLink anomalies				
Command injection				
Telemetry manipulation				
Denial of service				
Firmware integrity				

5.3 Required Outputs

Output	Required Content
Detection accuracy	Per-class and overall results with counts and confidence intervals, on data not used for tuning
Subtle-attack results	Detection against attack magnitude, including attacks close to the noise level
Detection latency	Distribution of time to first alert (median and 95th percentile) per class
Baseline comparison	The IDS against a simple threshold or the autopilot's own checks
Missed detections	Every missed attack listed and explained
Confidence statement	Why the team trusts the detection results, and what it did not test

Insert confusion matrix and per-class detection results here

Insert detection latency distribution and detection-versus-attack-magnitude plots here

6. Data Sources, Features and Detection Analysis [12 Marks]

Show which telemetry each detector depends on, and that every alert can be traced to the telemetry that caused it.

6.1 Data Sources and Features

Data source (message / field)	Feature derived	Used by detector	Why it matters	Real-data evidence

6.2 Detection Capabilities and Robustness

Capability	Method	Test evidence	Result	Limitation
GPS spoofing				
MAVLink anomalies (replay, sequence, rate)				
Command injection				
Telemetry manipulation				
Denial of service				
Firmware integrity				

Handling of malformed, truncated or corrupted telemetry				

6.3 Traceability of Alerts

For a sample of alerts, show the telemetry that caused each one.

Alert	Source message(s) and field values	Detector and threshold	How a reviewer checks it

Insert an example of alert evidence with the triggering telemetry here

7. False-Alarm Behaviour and Alerting [8 Marks]

Describe how the IDS behaves on normal flight and how alerts are raised and delivered. False-alarm results are only meaningful with enough varied benign flight, so state the exposure honestly.

Parameter	Value	Unit	Basis / Evidence
Benign flight time tested		h	

Benign distance tested (state the definition of 'distance covered' used)		km	
Number of separate benign flights		count	
False alarms observed		count	
False-alarm rate		per hour	
Upper 95% bound on false-alarm rate		per hour	
Alert persistence and suppression settings		--	
Alert response policy (advisory or automatic)		--	

7.1 Benign-Condition Matrix

Benign condition	Source (flight log, live telemetry)	Duration	Alerts raised	Assessment
Normal cruise				
Aggressive manoeuvres and mode changes				
Wind or turbulence				
GPS glitch or EKF reset (real)				

Telemetry link loss and recovery				
Take-off and landing				

7.2 Alert Workflow and Integrity

Alert Stage	Mechanism	Expected Behaviour	Evidence / Plot

Insert alert timeline from a benign flight and an attack flight here

Explain how alerts and logs are made tamper-evident, and what happens if the IDS itself fails.

8. Validation, Testing and Reproducibility [14 Marks]

The committee will re-run the submitted tests on a clean machine. Results that the submitted code and data cannot reproduce will not be credited.

8.1 Test Setup

Item	Details Required

Test framework and location	Framework, number of tests, and where they are in the submitted archive
Data used	Flight logs and live telemetry from actual flights, with any generated data labelled separately, and the source, size and archive location of each; large files included or supplied with a checksum
Data separation	Calibration, validation and test sets; how they were separated (by whole flight)
Attack generation	How attacks were created, by whom, and whether the detector could see how they were made
Ground truth	How the correct label for each test was established
Environment	Operating system, language and library versions, offline run
Independent re-run instructions	Exact commands a reviewer can run to reproduce the results

8.2 Results Summary

Test Category	Count	Passed	Failed / Skipped	Cause of Failures	Evidence
Unit tests					
Integration / end-to-end					
Replay of recorded flights					

Attack runs during flight operation					
Benign-flight runs					
Cross-platform (ArduPilot / PX4)					

8.3 Claims Traceability

List every quantitative claim in this report (accuracy, false-alarm rate, latency, resource use, tests) and show where it can be reproduced.

Claim in this report	Where reproduced (file / test)	Command to reproduce	Result

State the validation conclusion: which results are validated on real or independent data, which only on generated data, and which are not yet validated.

9. Compute, Integration, Usability and Reporting [8 Marks]

State where each measurement was taken. A measurement on a laptop or workstation must not be presented as a measurement on the target hardware.

Quantity	Value	Measured on (target / host)	Basis
CPU use at sustained telemetry rate			
Peak memory			
Per-message processing time (median, 99th percentile)			
Sustained messages per second			
Headroom at the maximum telemetry rate			
Power draw and added mass			

Item	Selected Approach	Evidence
Tap point and method (passive tap, router, onboard code)		
Changes required to the flight stack		
Protocol support (MAVLink 1 and 2, signing)		
Autopilot support (ArduPilot, PX4)		

Ground station and dashboard integration		
Behaviour if the IDS fails or is overloaded		
Installation and configuration steps		
Alert delivery and operator view		
Demonstration video (screen recording of a real run)		

Quantity	Value	Unit	Basis
Install time on a clean machine		min	
Time for an operator to interpret a typical alert		min	
Total cost of the recommended kit		INR	

Insert integration diagram and a screenshot of the operator view here

10. Final Deployable Package [7 Marks]

Required Item	Submitted? Yes / No	File Name / Section Reference
Complete source code (in the archive)		

Installation and user guide		
Test suite and instructions to run it		
Flight logs and datasets used, with licences		
Attack-injection scripts and ground-truth files		
Result files that reproduce the reported numbers		
Configuration files and thresholds		
Demonstration video (real run, not slides)		
Dependency list with licences; third-party code identified		
Risk register		
Known limitations and roadmap		

Readiness Item	Final Value / Status	Evidence
Detects all six attack classes in the team's tests?		
False-alarm rate within the team's stated target?		

Detection latency within the stated target?		
Compute within budget on the target hardware?		
Runs on an unmodified ArduPilot or PX4 stack?		
Alerts and logs tamper-evident?		
Third-party code identified and licensed?		
Operated in flight and ready for extended flight trials?		

Final readiness statement: Based on the validation completed, explain why this IDS should be selected for extended flight trials.

11. References, Data Traceability and Integrity [5 Marks]

No.	Reference / Source	Link / DOI / Datasheet	Application in the IDS
1			
2			
3			
4			

5			

11.1 Software, Digital and AI Tools Used

Tool	Purpose	Nature / Extent of Use	Team Verification Performed

11.2 Third-Party Code and Datasets

Name	Licence	What Was Used	Where Credited in this Report

11.3 Design Integrity Declaration

We confirm that the IDS design, source code, test cases, attack scenarios, results and technical decisions presented in this report have been reviewed and verified by the team. Every result shown was produced by the submitted code on the data described. No detector reads ground-truth labels or attack flags, and no result has been copied, estimated or edited to resemble a measurement unless it is labelled as such. The team accepts responsibility for the technical accuracy, validity and integrity of the submitted work. The evaluation committee may request the source repository history, test logs, raw flight data, or other technical records, and may require the team to explain, reproduce, modify or defend the technical work presented in this report.

Particular	Details
Team ID	
Team Name	
Team Leader Signature	
Date	

Stage 2 Evaluation Rubric

Criterion	Weight	Expectations
Executive IDS summary	5	Clear final IDS and key evidence; concise statement of confidence and limits
Stage 1 traceability and design evolution	8	Changes from Stage 1 are justified by testing, not arbitrary; Stage 1 claims are demonstrated or withdrawn
Updated requirements and architecture	7	Architecture is coherent, integrated and aligned with the challenge: detect the six attack classes in real time without disturbing flight
Integrated methodology and workflow	8	Logical workflow connecting data sources, features, detection, fusion, alerting and testing

Detection methodology and attack coverage	18	All six classes addressed; detectors decide from telemetry and do not read labels or attack flags; attacks include subtle cases, not only gross constants; per-class results with counts and confidence intervals on untuned data; latency distribution; baseline comparison; misses explained
Data sources, features and detection analysis	12	Each detector's inputs are listed and justified; every alert traces to the telemetry that caused it; malformed input handled safely; limits documented
False-alarm behaviour and alerting	8	Benign exposure of meaningful length and variety on actual flight operation; false-alarm rate per hour with an upper bound; clear alert policy that cannot interfere with flight; tamper-evident alerts and logs
Validation, testing and reproducibility	14	Tests and data are in the archive and pass on the committee's re-run; result files reproduce the reported numbers; no copied or placeholder results; claims traceable; conclusions clear
Compute, integration, usability and reporting	8	Resource use measured on the stated target hardware, with host measurements labelled as such; tap point, protocol and autopilot support are credible; behaviour when the IDS fails is safe; an operator can run it and read the alerts
Final deployable package	7	Source, guides, tests, datasets, injection scripts, result files, video and risk register are complete and run on a clean machine
References, traceability and integrity	5	Sources traceable; third-party code credited and licensed; tool usage (including AI tools) declared and verified; integrity declaration completed
Total	100	

Stage 2 Evaluation Philosophy

- Stage 2 focuses on a validated, working IDS and the evidence for it, not on a list of features.
- Teams are expected to improve substantially beyond the Stage 1 proof of concept.
- Validation effort should be significant but practical for a two-month team effort.
- A smaller IDS with traceable results and defensible claims is preferred over a larger one with unproven claims.
- Results on attacks that are separable by construction carry little weight; subtle attacks and honest benign data carry the most.
- Evidence from actual flight operation carries full weight; results from simulated or generated data are treated as design evidence and carry limited weight.
- Anything that is not demonstrated is treated as design, not as capability.
- Results that the submitted code and data cannot reproduce on the committee's re-run will not be credited.
- The final output should have the IDS which can proceed to extended flight trials.